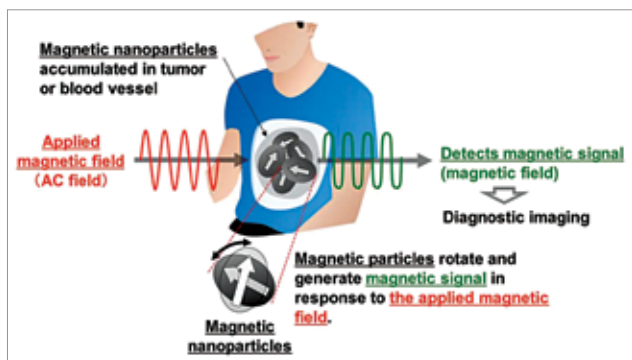


High-sensitivity magnetic sensor for better image diagnosis



High-sensitivity magnetic sensor (Credit: www.tdk.com/en)

Yokohama National University (YNU) and TDK Corporation have developed a prototype image diagnosis technology utilising a high-sensitivity magnetic sensor. The developed prototype technology is related to the magnetic particle imaging method, which is intended to detect and create images of magnetic particles accumulated in a tumor or blood vessel. When intended for use in medical imaging, it is important for devices to be highly sensitive to detect small amounts of magnetic particles. Though magnetic particle imaging technologies primarily use a method that measures electromotive force electromagnetically induced through detection coils, the new technology developed by YNU utilises a prototype high-sensitivity magnetic sensor to achieve this.

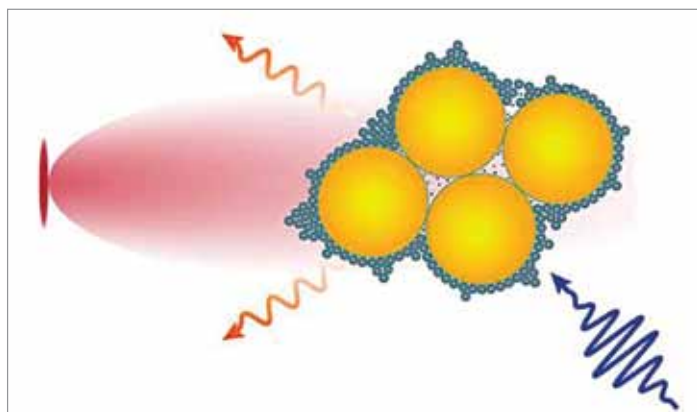
Smart clothing that incorporates conductive fibres

There's no need to wear uncomfortable smartwatches or chest straps to monitor your heart if your comfy shirt can do a better job. That's the idea behind 'smart clothing' developed by the Rice University lab, which employed conductive nanotube thread to help enable heart rate monitoring and perform continuous electrocardiogram (EKG) of the wearer. The fibres are just as conductive as metal wires, but washable, comfortable, and far less likely to break when a body is in motion, according to the researchers. On the whole, the enhancement can gather data better than a standard chest-strap monitor taking live measurements. When matched with commercial medical electrode monitors, the carbon nanotube shirt gives slightly better EKGs.

Lauren Taylor from Rice University shows a shirt with carbon nanotube thread that provides constant monitoring of the wearer's heart (Credit: <https://news.rice.edu>)



Nano-camera for real-time monitoring of chemical reactions



Researchers have made a tiny camera, held together with 'molecular glue' that allows them to observe chemical reactions in real time (Credit: www.cam.ac.uk/research)

Researchers at the University of Cambridge have made a tiny camera that is held together with a 'molecular glue,' allowing them to observe chemical reactions in real time. The device combines tiny semiconductor nanocrystals called quantum dots and gold nanoparticles using molecular glue called cucurbituril (CB). When added to water with the molecule to be studied, the components self-assemble in seconds into a stable, powerful tool that allows the real-time monitoring of chemical reactions. The camera harvests light within the semiconductors, inducing electron transfer processes like those that occur in photosynthesis, which can be monitored using incorporated gold nanoparticle sensors and spectroscopic techniques.